GASTÓN ECHENIQUE

Senior Data Engineer · AWS & Databricks · Lakehouses & Data Pipelines

Buenos Aires, Argentina (remote)

LinkedIn · GitHub · Portfolio

Contact: LinkedIn or the contact form on my website

SUMMARY

Senior Data Engineer with 7+ years in data, building production lakehouses and data pipelines on AWS (Glue/PySpark, Athena, Lambda, S3, Iceberg), plus a hands-on Databricks portfolio (Unity Catalog, Lakeflow, Terraform, CI/CD). I own the full path: ingestion from APIs and operational databases, RAW/STAGING/CONSUME modeling, idempotent config-driven pipelines, data quality and delivery to product teams. My main 2026 work is a healthcare-professional verification product covering three countries, running in production for a LATAM digital health platform. Background in hydrographic and geospatial data (Argentine Navy; Project Mineral at X, the moonshot factory). Open to freelance and contract projects.

Core stack: AWS (S3, Glue/PySpark, Athena, Lambda, RDS, DynamoDB, SSM, Bedrock) · Databricks (Unity Catalog, Lakeflow, DABs) · Apache Iceberg · dbt · Airflow · Terraform · SQL · Python · Azure (ADF, Synapse) · GCP (BigQuery)

PROFESSIONAL EXPERIENCE

Senior Data Engineer — The Black Puma AI (AWS Partner consultancy)

Sep 2022 – Present · Buenos Aires (remote)

- **Healthcare-professional verification (Argentina, Brazil, Colombia):** built the pipelines on AWS (Glue/PySpark, Athena) that match platform users against official professional registries and third-party APIs.
  - Reprocessed 115,796 records.
  - Assigned 600,000+ "verified professional" badges in bulk, with a 79.3% automatic match rate.
  - Analyzed a 102,508-user universe for the third country.
- **Lakehouse on AWS (Glue + Apache Iceberg, RAW/STAGING/CONSUME):** integrated 8 new data sources (official registries, third-party REST APIs, GA4/Firebase via BigQuery, LMS on MariaDB, operational RDS databases) and delivered 6 data products to product and marketing teams.
- **Platform modernization:**
  - migrated 8 legacy jobs to a parameterized, idempotent, config-driven architecture (AWS SSM Parameter Store) with no hardcoded settings;
  - refactored repetitive loads into a generic RDS ingestion job;
  - upgraded to AWS Glue 6.0;
  - moved the SendGrid consume layer to incremental Parquet + MERGE.
- **Reliability:** resolved schema drift in Athena partitions, silent JDBC/SSL failures and source-unavailability issues in the BigQuery → S3 pipeline.
- **Pharma data lakes:** maintained AWS data lakes for two pharmaceutical companies, with monthly data-quality and integrity reports, SFTP feed monitoring and log-based pipeline monitoring.
- **Earlier engagements:** Azure Data Factory + Synapse pipelines for multi-country financial and operational data; H3-based geomarketing combining POI and customer data; deployment of a fraud-detection model to AWS SageMaker for an energy utility; data-ops automation with n8n + Docker.
- Designed a Databricks lakehouse architecture proposal for a client (pre-sales).
- Onboarded and transferred knowledge to new data engineers on the team.

Data Engineer (Contractor) — X, the moonshot factory (Alphabet) · Project Mineral

Aug 2021 – Aug 2022 · Remote

- Rebuilt the data foundation for an agricultural analytics effort, turning 4 years of disconnected ground-truth and GIS datasets into production-ready pipelines feeding crop-yield prediction models.
- Designed GCP data-lake ingestion (BigQuery, Dataflow) with YAML-driven configuration for reliable, repeatable loads.
- Built automated data-quality validation into multi-year historical ETL, plus interactive visualizations for data science and partner teams.
- *Project Mineral has since graduated as a Google company.*

Argentine Navy — Hydrography, Nautical Cartography & Oceanographic Data

2010 – Jun 2021

- **Cartographer, Naval Hydrographic Service (2019–2021):** migrated nautical chart data from S-57 ENC toward the data-driven S-100 family (structured databases, XML/GML) and built web mapping with PostGIS, GeoServer and OpenLayers. This was my entry point into data engineering.

- **Captain, hydrographic vessel ARA "Cormorán" (2016–2018):** led a 20+ person multidisciplinary team, implemented data management for hydrographic operations, and ran public procurement.
- **Oceanographic data analyst (2018); Head of Administration & Finance (2013–2015):** statistical analysis of oceanographic data, management KPIs and budget planning.
- NASA Space Apps Challenge 2019 — "Galactic Problem-Solver" award (team lead).

Teaching (2024 – 2025)

- **Python & Deep Learning for Remote Sensing Instructor** — GeoDojo (formerly educacion.earth) · 2024 – 2025
- **AI & Machine Learning Instructor** — Universidad Nacional de Tres de Febrero (UNTREF) · 2025

PROJECTS

- **SEC EDGAR Analytics Agent** — *UdeSA Agentic AI Executive Program, AWS-sponsored in-person hackathon, Jun 2026.*
  - End-to-end text-to-SQL agent on AWS: natural-language question → SQL → Athena over public SEC EDGAR data in S3 → natural-language answer.
  - 3-layer architecture (tools → sub-agents → orchestrator) with Strands Agents SDK and Amazon Bedrock (Claude Haiku), plus persistent multi-session memory in DynamoDB.
  - Streamlit front end via Cloudflare Tunnel. MVP built in one day; total infrastructure cost: USD 0.26.
  - Designed to be re-pointed to a client's own tables to replace dashboards with natural-language queries.
- **Entity 360 — multicloud golden record on Databricks.**
  - Resolves 5 real sources from 4 producers (legacy SQL Server with CDC, SEC via AWS Lambda, GDELT on BigQuery, OpenSanctions/Wikidata) into an explainable golden record.
  - Data contracts with quarantine; Gold built with dbt on Iceberg and read by Snowflake zero-copy; orchestrated end-to-end with Airflow.
  - Total cost under USD 1.
- **Pokédex Agent — data platform + Claude agent on Databricks.**
  - Terraform-provisioned workspace with Unity Catalog and a medallion pipeline gated by data-quality checks.
  - A native MCP server runs as a Databricks App, and a multi-agent coordinator answers only from curated Gold data.
- **Medallion pipeline on Databricks:** Terraform + CI/CD with dev/prod promotion. Forked by other engineers.
- **Real-time satellite streaming:** SGP4 propagation → Azure Event Hubs → Databricks Structured Streaming, feeding a live medallion pipeline and dashboard.
- **Lakebase multi-format:** ingests 5 formats into Databricks Lakebase (managed Postgres), queryable live from Unity Catalog via Lakehouse Federation.
- **Tourism advisor on Azure Databricks:** multi-source medallion (5 sources) plus a tool-calling agent, MCP server and map/chat front end, with Azure infra via Terraform.

Code and videos: github.com/oechenique

CERTIFICATIONS & CREDENTIALS

- **AWS Certified AI Practitioner (AIF-C01)** — 2025
- **AWS microcredentials:** Agentic AI Demonstrated · Data Lakehouse Demonstrated — 2026
- **Databricks Academy, 19 badges & accreditations (2025–2026):**
  - Platform Architect (AWS, Azure, GCP) · Platform Administrator
  - Lakeflow Spark Declarative Pipelines (build + advanced) · Lakeflow Connect · Declarative Automation Bundles · DevOps Essentials for Data Engineering · Data Privacy
  - ML Model Development & Deployment · ML at Scale · AI Agent Fundamentals
- **Anthropic Academy (2026):** Building with the Claude API · Claude Code in Action · Model Context Protocol · Agent Skills

EDUCATION

- **Executive Education Program — Agentic AI**, Universidad de San Andrés (UdeSA) — Apr–Jun 2026
- **Postgraduate Diploma in Big Data and Spatial Territorial Intelligence**, FLACSO — 2022
- **Data Science program**, Digital House — 2021–2022
- **MSc in Environmental Management**, UNSAM — 2017
- **Specialization in Hydrography & Oceanography**, Argentine Navy Faculty (Facultad de la Armada) — 2015–2016
- **BA in Naval Resource Management**, Argentine Naval Academy (Escuela Naval Militar) / UNDEF — 2005–2009

Languages: Spanish (native) · English (full professional)